



Intelligence Academy

Scikit-Learn 101

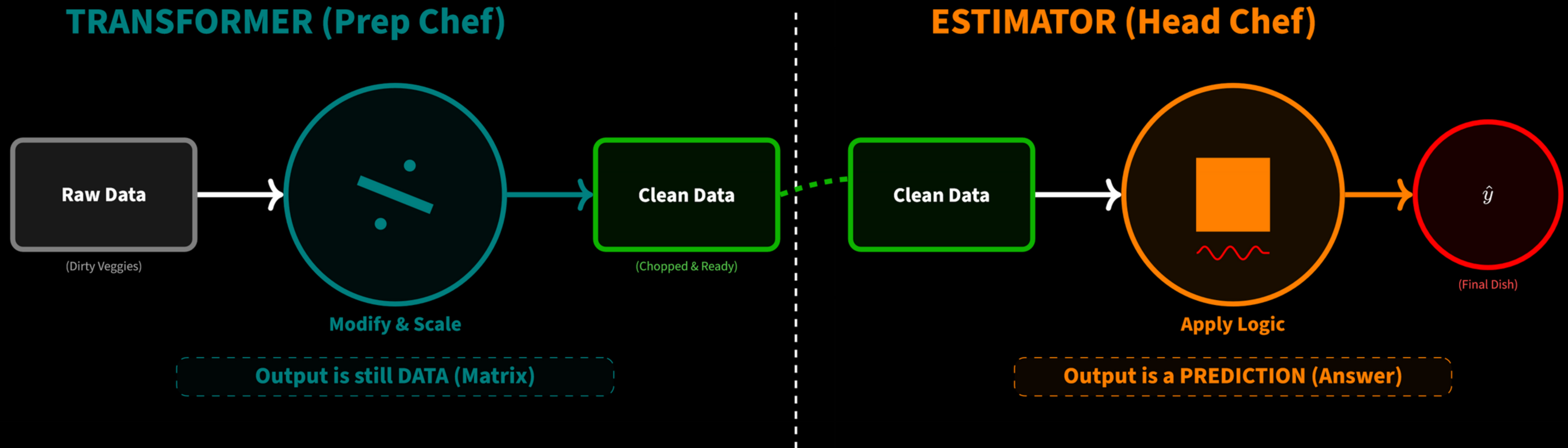
05. Transformers vs Estimators

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5.1 The Fundamental Split: Purpose & Role

The easiest way to understand the difference is through a kitchen analogy.

- **Transformers (Prep Chefs):** Their job is to **modify** the raw ingredients—cleaning, chopping, and scaling them for the next step.
- **Estimators (Head Chefs):** Their job is to **consume** the prepared ingredients to cook the final dish (make the prediction).

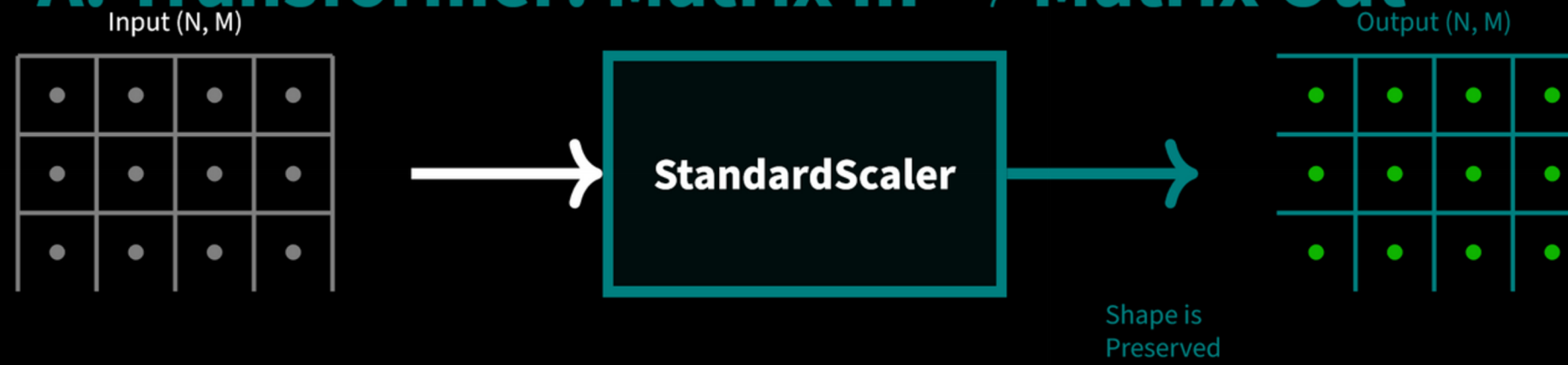


5.2 The Shape Logic: Matrix vs. Vector

You can identify them by checking the **Output Shape**.

- **Transformers (Consistency):** They input a Matrix and output a Matrix. The data structure is preserved for the next step.
- **Estimators (Collapse):** They input a Matrix but output a Vector. The complex data is collapsed into a single list of answers.

A. Transformer: Matrix In $\rightarrow$ Matrix Out



B. Estimator: Matrix In $\rightarrow$ Vector Out



5.3 The API Signature: Methods Checklist

In Scikit-Learn, an object's type is defined by the methods it implements.

- **Transformer:** Must have `.transform()`. Usually has `.fit_transform()`.
- **Estimator:** Must have `.predict()`. Usually has `.score()`.

Class Transformer

```
def fit(X, y=None): ...  
  
def transform(X): ... ← The Modifier  
  
def fit_transform(X): ... ← The Shortcut  
  
# NO predict()
```

Class Estimator

```
def fit(X, y): ...  
  
def predict(X): ... ← The Answer  
  
def score(X, y): ... ← The Grade  
  
# NO transform()
```